```
import java.util.Scanner;


public class railfence3a {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);


        System.out.print("Enter message: ");
        String text = sc.nextLine()
                    .toUpperCase()
                    .replace(" ", "");


        String rail1 = "";
        String rail2 = "";
// Encryption
        for (int i = 0; i < text.length(); i++) {
            if (i % 2 == 0)
                rail1 += text.charAt(i);
            else
                rail2 += text.charAt(i);
        }
        System.out.println("\nRail arrangement:");
```

```
// Display Rail 1
for (int i = 0; i < text.length(); i++) {
    if (i % 2 == 0)
        System.out.print(text.charAt(i) + " ");
    else
        System.out.print("  ");
}

System.out.println();

// Display Rail 2
for (int i = 0; i < text.length(); i++) {
    if (i % 2 != 0)
        System.out.print(text.charAt(i) + " ");
    else
        System.out.print("  ");
}

String encrypted = rail1 + rail2;

System.out.println("\n\nRail 1: " + rail1);
```

```
        System.out.println("Rail 2: " + rail2);
        System.out.println("Encrypted: " + encrypted);

        // Decryption
        int middle = (text.length() + 1) / 2;
        rail1 = encrypted.substring(0, middle);
        rail2 = encrypted.substring(middle);

        String decrypted = "";

        for (int i = 0; i < middle; i++) {
            decrypted += rail1.charAt(i);

            if (i < rail2.length())
                decrypted += rail2.charAt(i);
        }

        System.out.println("Decrypted: " + decrypted);
        sc.close();
    }
}
```